

Architecture change

Phase I Mastermind extended to Wordle | Source of truth: submitted Phase I code and Part II specification.

01 Reused unchanged

Code, GameConfiguration, SecretGenerator, FixedSecretGenerator, RandomSecretGenerator, GuessValidator, DefaultGuessValidator, ValidationResult, GuessEvaluator, Feedback, GuessRecord, UserInterface, MastermindConfiguration, MastermindEvaluator, and MastermindFeedback. RandomSecretGenerator and DefaultGuessValidator remain Mastermind tools; they cannot generate or validate whole dictionary words.

02 Existing types modified

Type	Change and reason
Driver	Added the Wordle launch and external dictionary path.
MastermindGame	Delegates replay to GameSession; retains Mastermind welcome and secret prompts.
GuessingGameEngine	Accepts a word/code label while preserving the old constructor; clears history per new round and handles input ending.
GuessHistory	Added clear() for an engine reused across rounds.
ConsoleUserInterface	Signals exhausted input with null to avoid an endless invalid-guess loop.

03 New types

WordleConfiguration loads the word file. WordleValidator checks length, letters and dictionary membership. WordleSecretGenerator chooses a complete listed word. WordleEvaluator and WordleFeedback produce per-position C/P/A results. WordleGame supplies the welcome and test-mode word choice. Shared GameSession coordinates replay; Phase2Tests holds the automated checks.

Design decisions

04 Where Phase I was too specific

RandomSecretGenerator samples individual symbols, which rarely yields a real word. DefaultGuessValidator accepts any length-correct sequence from an alphabet, including nonsense words. GuessingGameEngine assumed the secret should always be called a code, and MastermindGame owned a replay loop that both games need.

05 The refactoring

GameSession owns one replay prompt and creates one GuessingGameEngine per round. Both games inject their own configuration, validator, evaluator and secret-generator supplier. The shared engine still owns attempts, HISTORY, test-only REVEAL, result reporting and history.

06 Complexity and value

The engine gained a secretName label and a backwards-compatible overloaded constructor so messages say code for Mastermind and word for Wordle. One coordinator class is the cost of sharing replay. Dictionary validation and full-word selection stay in Wordle-specific types; neither interface needed a breaking change.

07 If Wordle had been known in Phase I

I would have moved replay into a shared session coordinator immediately, made secret terminology configurable, and described alphabet membership as only the simplest kind of validation. The existing GuessValidator contract was broad enough to accept a dictionary validator.

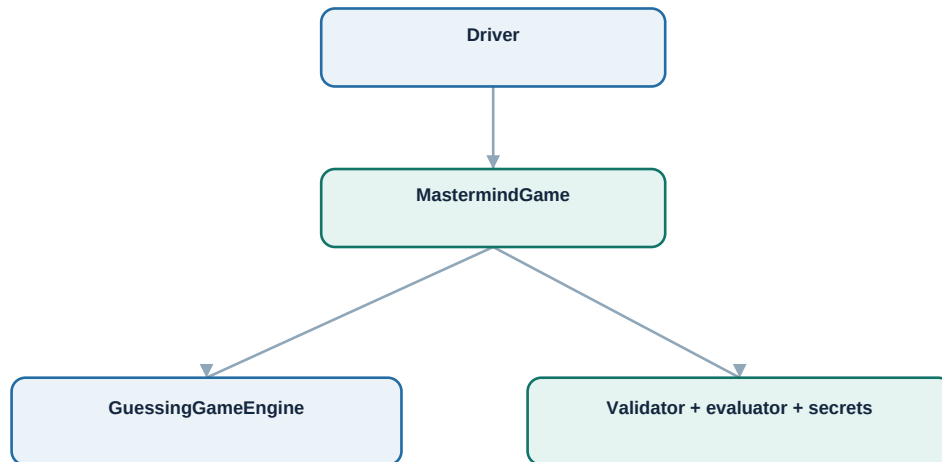
Data and scope

The 534-entry list is external at data/words.txt. Both guesses and secrets come from it. The original Design.pdf remains as the Phase I baseline; the new diagrams on the next page show the change.

Before and after

Arrows mean “uses.” Boxes are grouped by shared infrastructure (blue), game assembly (green), and data (amber). The diagrams show the main control path; additional unchanged interfaces are listed on page 1.

Before / Phase I



After / Phase II

